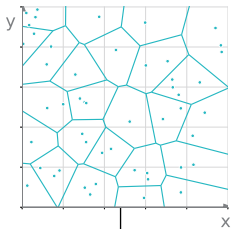


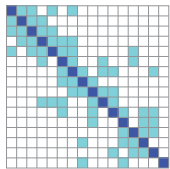
GeoDCD: 几何动态因果发现

时空序列输入
 $X \in \mathbb{R}^{N \times T}$

几何池化器 (物理空间)



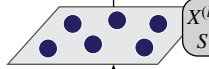
分配矩阵 S



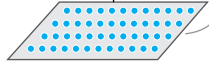
级别2
(粗糙)



级别1
(中等)



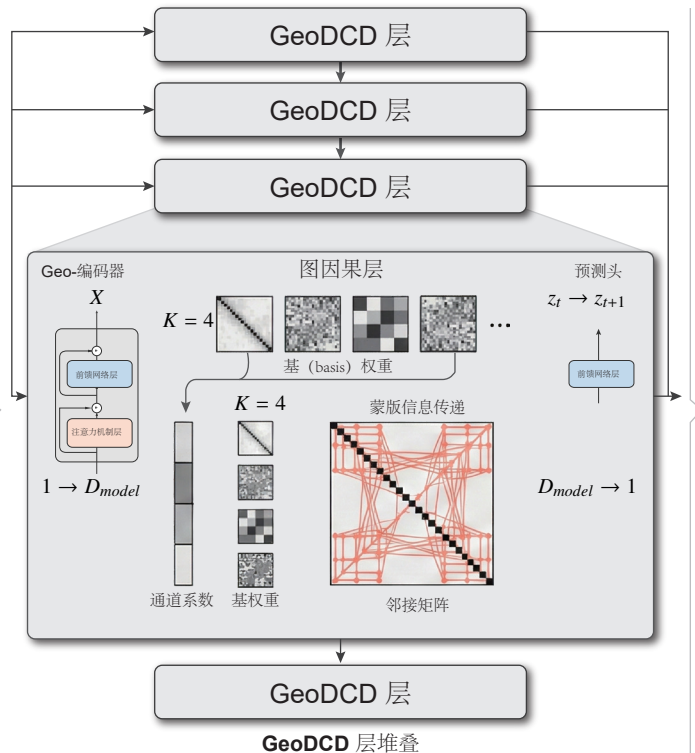
级别0
(精细)



$$X^{(l+1)} = S^T X^{(l)}$$

层次化金字塔结构

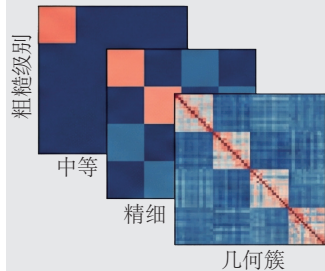
X



$\mathcal{L}_{pred}$ 预测(Prediction)损失

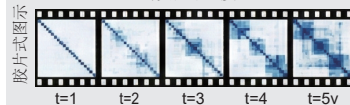
$\mathcal{L}_{structural}$ L1稀疏(L1 Sparsity)损失

输出
静态结构



动态演化

时变因果强度



通过几何基分解的无监督发现